

PROJECT REPORT

Workforce Attrition Patterns and Risk Hotspot Analysis at Palo Alto Networks

Project Type	Technical Documentation / Data Analyst Internship Project
Organization	Palo Alto Networks
Developed By	Taniya
Role	Data Analyst Intern
Technology Stack	Python, Pandas, Streamlit, Plotly

Purpose: To analyze employee attrition patterns, identify workforce risk hotspots, and present actionable HR insights through an interactive analytics dashboard.

1. Introduction

Employee attrition is an important workforce-management challenge, particularly in a high-skill and fast-paced technology organization. Employee exits can contribute to loss of critical technical expertise, increased hiring and onboarding costs, disruption to long-term projects, and potential effects on employee morale and productivity.

This project develops an interactive HR analytics dashboard focused on understanding employee attrition patterns and identifying segments where attrition risk may be comparatively higher. The dashboard converts employee-level HR data into visual metrics, interactive charts, filterable views, key insights, and business-oriented recommendations.

2. Project Objectives

- Analyze overall employee attrition and retention patterns.
- Identify departments and job roles with comparatively higher attrition rates.
- Study attrition across age and gender groups.
- Examine education and education-field patterns.
- Analyze marital-status related attrition patterns.
- Study tenure, years since last promotion, and career-stage patterns.
- Assess overtime and business-travel patterns in relation to attrition.
- Explore distance-from-home, job satisfaction, and monthly-income patterns.
- Differentiate early-tenure employees from more experienced employees.
- Present findings through an interactive dashboard to support data-driven HR decision-making.

3. Business Problem

The project addresses the need to understand where and among whom employee attrition is most visible. Instead of relying only on an overall attrition percentage, the analysis examines multiple workforce dimensions so that potential risk hotspots can be viewed at department, role, demographic, tenure, workload, mobility, and compensation levels.

4. Dataset and Data Preparation

The dashboard uses the **Palo Alto Networks.csv** employee HR dataset. The data is loaded with Pandas and filtered dynamically according to the selections made in the dashboard sidebar.

The dashboard derives analytical labels and measures required for visualization, including attrition labels, attrition rates, tenure buckets, career-stage groupings, and other calculated metrics. Filtering is applied before the dashboard metrics and charts are calculated so that the displayed results respond to the user's selections.

5. Tools and Technologies

Technology	Use in the Project
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Python	Core programming, calculations, and dashboard logic.
Pandas	Dataset loading, filtering, transformation, grouping, and analysis.
Streamlit	Interactive web dashboard and user interface.
Plotly	Interactive charts and data visualizations.
VS Code	Development and code-editing environment.

6. Methodology

The project follows an exploratory HR analytics workflow. First, the employee dataset is loaded and prepared. Next, descriptive measures and grouped attrition rates are calculated. The analysis is then presented through interactive visualizations. Finally, the observed patterns are translated into key insights and business recommendations.

- **Data Loading:** Import the employee dataset using Pandas.
- **Data Preparation:** Create analytical labels and calculated measures required by the dashboard.
- **Filtering:** Allow users to select Department, Job Role, Overtime, and Business Travel.
- **Descriptive Analysis:** Calculate employee counts, exited employees, retained employees, and attrition rates.
- **Comparative Analysis:** Compare attrition across demographic, career, workload, mobility, and compensation dimensions.
- **Visualization:** Present results through interactive Plotly charts and Streamlit components.
- **Insight Generation:** Highlight high-risk areas and provide business-oriented recommendations.

7. Dashboard Design and Features

The dashboard is designed as an interactive HR analytics interface. A sidebar provides filters, while the main area presents KPI cards, charts, employee profiles, analysis sections, and dynamic business insights.

- **KPI Overview:** Total Employees, Attrition Rate, Employees Exited, Employees Retained, and Early-Tenure Attrition.
- **Attrition Distribution:** Visual comparison of retained and exited employees.
- **Exited Employee Profile:** Average age, tenure, distance from home, and monthly income for exited employees.
- **Department & Role Analysis:** Department-wise and job-role-wise attrition rates.
- **High-Risk Areas:** Identification of areas showing comparatively elevated attrition.
- **Demographic Explorer:** Age and gender analysis.
- **Education Analysis:** Education level and education-field analysis.
- **Marital Status:** Attrition comparison across marital-status groups.
- **Tenure & Career Stage:** Tenure buckets, years since last promotion, and career-stage analysis.

- **Workload & Mobility:** Overtime, business travel, and distance-from-home analysis.
- **Experience & Compensation:** Job satisfaction and monthly-income analysis.
- **Additional Indicators:** Overtime attrition, frequent-travel attrition, and exited-employee average tenure indicators.
- **Key Insights & Recommendations:** Dynamic business-focused observations based on the filtered data.

8. Key Analytical Questions

- Which departments have the highest attrition rates?
- Which job roles represent higher attrition risk?
- Which age groups show stronger attrition patterns?
- How does attrition vary by employee tenure?
- Are early-tenure employees more exposed to attrition?
- How is overtime associated with employee exits?
- How does business travel relate to attrition?
- Does distance from home show a meaningful attrition pattern?
- How do employee satisfaction and monthly income vary across attrition outcomes?

9. Results and Interpretation

The dashboard provides a consolidated view of employee attrition and allows the user to move from overall workforce metrics to more detailed risk analysis. Because the dashboard is filter-driven, the displayed KPIs and visualizations can change according to the selected department, job role, overtime status, and business-travel category.

In the unfiltered dashboard view used during development, the workforce contained **1,470 employees**, with **237 employees exited** and **1,233 employees retained**. The overall attrition rate displayed was approximately **16.12%**. These figures establish the baseline view before applying interactive filters.

10. Business Insights and Recommendations

The dashboard's Key Insights section is designed to translate analytical results into business-oriented observations. Rather than repeating KPI values, the section focuses on risk hotspots and potential retention actions.

- **Department Risk:** Pay closer attention to departments showing comparatively high attrition rates and investigate role-specific causes.
- **Job Role Risk:** Review retention challenges in roles with elevated employee exits, especially where specialized skills are difficult to replace.
- **Overtime Risk:** Monitor workload and overtime patterns and consider workload balancing, staffing, and employee well-being initiatives.
- **Early-Tenure Risk:** Strengthen onboarding, mentoring, early-career engagement, and manager support for employees in the first few years.

- **Travel Risk:** Review travel intensity and mobility expectations for employee groups showing higher attrition.
- **Retention Strategy:** Use the dashboard to prioritize deeper investigation and targeted retention actions rather than applying one strategy to the entire workforce.

11. Technical Implementation

The application is implemented as a Streamlit Python script named **Hr_dashboard.py**. The dashboard reads the CSV dataset, creates derived analytical fields, applies sidebar filters, calculates KPIs, and generates interactive Plotly visualizations. Custom styling is applied to improve readability and provide a professional dashboard appearance.

The project dependencies are documented in **requirements.txt**. The primary packages are Streamlit, Pandas, and Plotly.

12. Execution

Install the required packages using:

```
pip install -r requirements.txt
```

Run the dashboard using:

```
python -m streamlit run Hr_dashboard.py
```

13. Project Structure

Workforce-Attrition-Analysis/

■■■ Hr_dashboard.py

■■■ Palo Alto Networks.csv

■■■ requirements.txt

■■■ README.md

14. Conclusion

The Workforce Attrition Patterns and Risk Hotspot Analysis dashboard demonstrates how HR data can be transformed into an interactive analytical solution. By combining KPI-based summaries, comparative visualizations, interactive filters, and business recommendations, the project provides a structured way to explore employee attrition across multiple workforce dimensions.

The project also demonstrates practical Data Analyst skills including data preparation, exploratory analysis, grouping and aggregation, interactive visualization, dashboard design, and communication of business insights.

15. Future Scope

- Add additional HR datasets or historical periods for trend analysis.
- Introduce predictive attrition modeling for employee-level risk scoring.
- Add automated monitoring of attrition KPIs over time.
- Integrate additional employee-engagement or performance indicators where available.

- Deploy the dashboard as a public or organization-accessible web application.

16. Author

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